5 (a) Define a magnetic field.

A magnetic field is the region of space where a magnetic pole, moving charged particle or current-carrying conductor will experience a magnetic force.

Need all 3 components to award 2 marks
"region of space" and "magnetic force" tied together

…………[2]

(a) A 'bus bar' is a metal bar which can used to conduct a large electric current. In a test, two bus bars, X and Y, of length 0.90 m are clamped at either end parallel to each other on a base board, as shown in Fig. 5.1.

When a constant current of 12 kA is carried by each bus bar, they exert a force of 200 N on each other.

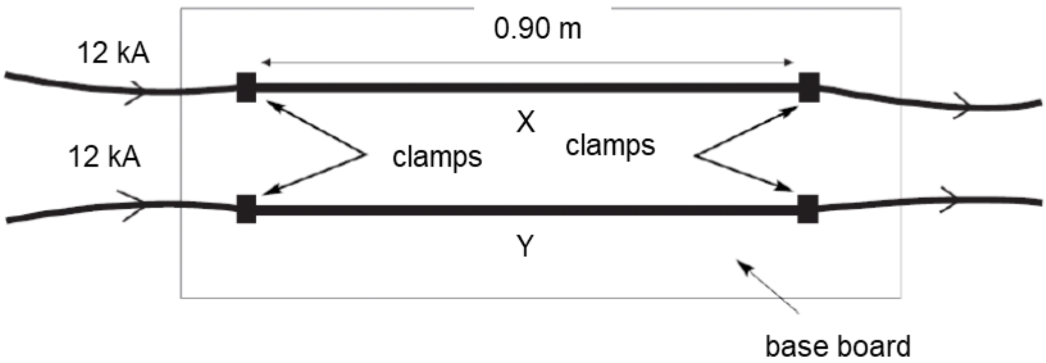

Fig. 5.1

(i) Calculate the magnetic flux density due to the current in one bus bar at the position of the other bus bar.

$$B = \frac{F}{IL} = \frac{200}{12\times10^{3}\times0.90}$$
$= 1.85 \times 10^{-2}$ T

magnetic flux density = ………………… T [2]

(ii) Calculate the magnitude of the force on each bus bar if X carried a current of 6.0 kA and Y carried a current of 12 kA in the same direction.

Force on X due to Y
B at X (due to Y) is unchanged
F_X = BIL and I is reduced to 6 kA
So F_X = 100 N
Force on Y due to X
B at Y (due to X) is reduced to 9.26×10^{-3} T
F_Y = BIL and I is unchanged
So F_Y is reduced to 100 N

magnetic force on X = ………………… N

magnetic force on Y = ………………… N [2]

(b) A small circular coil of cross sectional area 1.7×10^{-4} m^2 contains 250 turns of wire. The plane of the coil is placed parallel to and a distance x from the pole of a magnet as shown in Fig. 5.2.

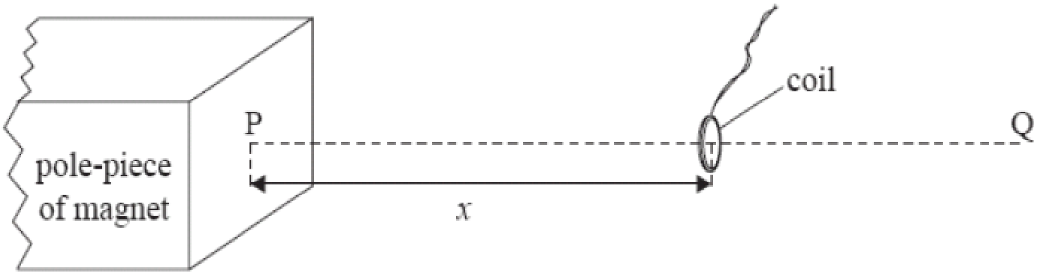

Fig. 5.2

PQ is a line that is normal to the pole piece. The variation with distance *x* along line PQ of the mean magnetic flux density *B* in the coil is shown in Fig. 5.3.

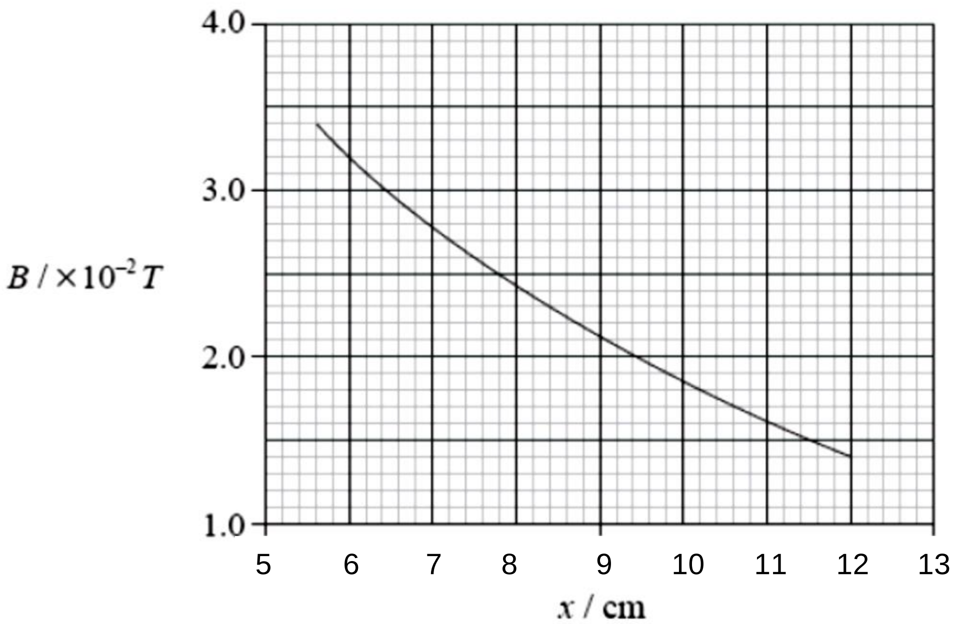

Fig. 5.3

(i) For the coil situated a distance 6.0 cm from the pole piece of the magnet,

1. State the average magnetic flux density in the coil.

average magnetic flux density = 3.2×10^{-2} T ……… T [1]

2. Calculate the flux linkage through the coil.

flux linkage = NBA = $250 \times 3.2 \times 10^{-2} \times 1.7 \times 10^{-4}$

= $1.36 \times 10^{-3} \sim 1.4 \times 10^{-3}$ Wb-turns magnetic flux linkage = ………………… Wb-turns [2]

(ii) The coil is moved along PQ so that the distance *x* changes from 6.0 cm to 12.0 cm in a time of 0.35 s.

1. Determine the change in magnetic flux linkage through the coil.

From the graph, ΔB = $3.2 \times 10^{-2} - 1.4 \times 10^{-2} = 1.8 \times 10^{-2}$ T
therefore change in flux linkage = NBA = $250 \times 1.8 \times 10^{-2} \times 1.7 \times 10^{-4} = 7.65 \times 10^{-4} \sim 7.7 \times 10^{-4}$ Wb-turns
ecf from (i)

change in magnetic flux linkage = ………………… Wb-turns [2]

2. State Faraday's law of electromagnetic induction and hence calculate the mean e.m.f. induced in the coil.

e.m.f. is proportional to rate of change of flux <u>linkage</u>

$\varepsilon = \frac{(6.8\times10^{-4})}{0.35} = 2.19 \times 10^{-3}$

mean e.m.f. induced = ………………… V [2]

(iii) Use Lenz's law to explain why work has to be done to move the coil along the line PQ.

Induced current produces a magnetic field in the coil.
This produces a force.
The force acts to oppose the motion of the coil.

………………………………………………………………………… [2]

6 The isotope Iron-59 is a β-emitter with a half-life of 45 days. In order to estimate engine